

Chapter 2. Quality of Service in Telecommunications

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1. Conceptions in Quality of Service

How happy the telephone company is keeping the customer

- Can connectivity be achieved?
- Delay before receiving dial tone (*dial tone delay*).
- Post dial(ing) delay (*time from the completion of dialing the last digit of a number to the first ring-back 13 of the called telephone*). This is the primary measure of signaling quality.
- Availability of service tones (*e.g., busy tone, telephone out of order, time out, and all trunks busy (ATB)*).
- Correctness of billing.
- Reasonable cost of service to the customer.
- Responsiveness to servicing requests.
- Responsiveness and courtesy of operators.

Time to installation of a new telephone, and, by some,...

2. Measurement of QoS

Service of telecommunication: voice, data, and image

QoS for telecom:

- a) SNR for modulation, radio transmission
- b) Voice transmission
- c) Data transmission
- d) Video transmission

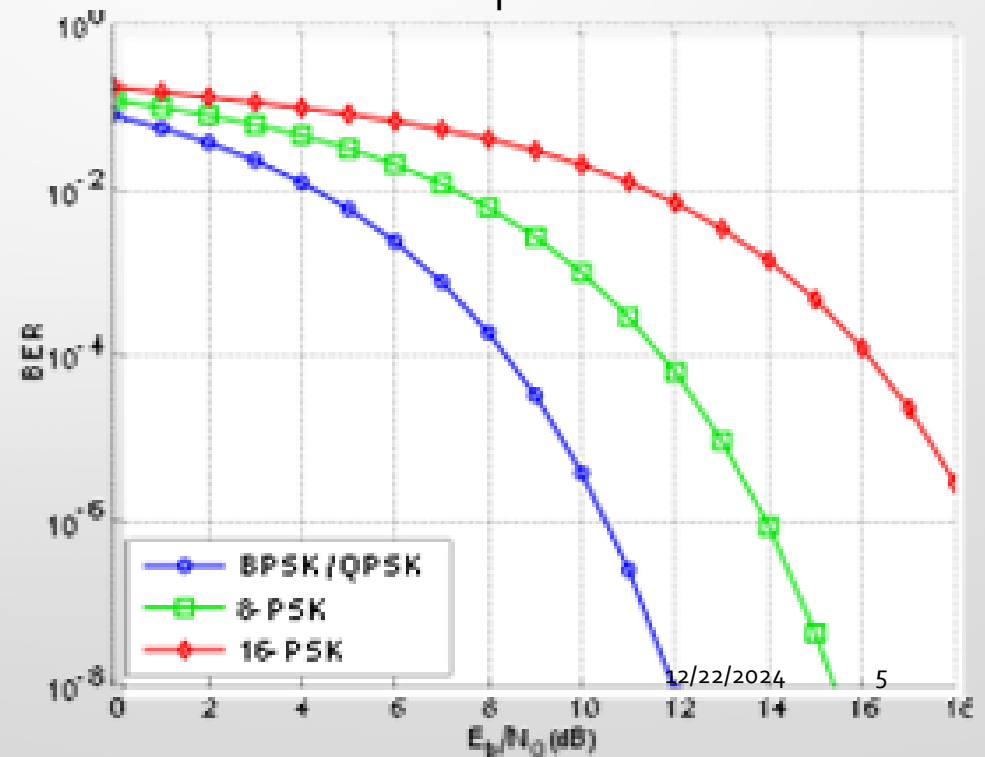
a) Signal-to-Noise ratio

S/N requirement at receivers:

- Voice: 40 dB
- Video (TV): 45 dB
- Data: 15dB (depend on modulation type and specified error performance)

$$SNR = \frac{S}{N}$$

Examples



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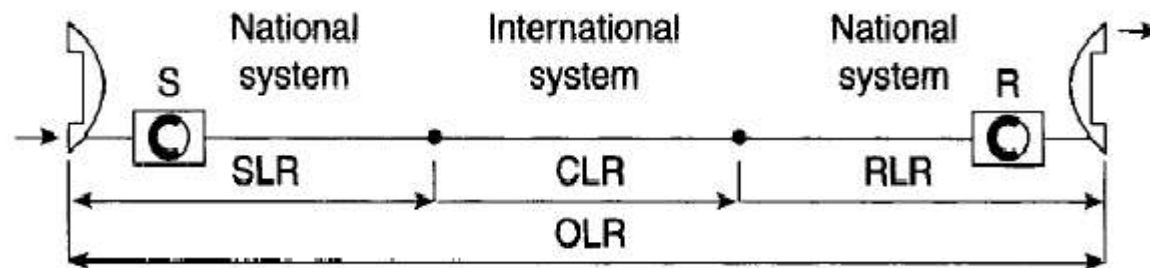
b) Voice transmission

Table 3.1 Overall Loudness Rating Opinion Results

Overall Loudness Rating (dB)	Representative Opinion Results ^a	
	Percent "Good plus Excellent"	Percent "Poor plus Bad"
5-15	<90	<1
20	80	4
25	65	10
30	45	20

^aBased on opinion relationship derived from the transmission quality index (see Annex A, ITU-T Rec. P.11).

Source: ITU-T Rec. P.11, Table 1/P.11, p. 2, Helsinki, 3/93.



CLR Circuit loudness rating

c) Data transmission

- In digital transmission, the number of **bit errors** is the number of received bits of a data stream over a communication channel that have been altered due to noise, interference, distortion or bit synchronization errors.
- The **bit error rate** or **bit error ratio (BER)** is the number of bit errors divided by the total number of transferred bits during a studied time interval. BER is a unitless performance measure, often expressed as a percentage.
- The standard BER for data transmission: $10^{-6} - 10^{-10}$

d) S/N of Video (television) standards

TASO stands for Television Allocations Study Organization

Quality		S/N
1.	Excellent (no perceptible snow)	45 dB
2.	Fine (snow just perceptible)	35 dB
3.	Passable (snow definitely perceptible but not objectionable)	29 dB
4.	Marginal (snow somewhat objectionable)	25 dB

CCIR FIVE-GRADE SCALE

Quality	Impairment
5. Excellent	5. Imperceptible
4. Good	4. Perceptible, but not annoying
3. Fair	3. Slightly annoying
2. Poor	2. Annoying
1. Bad	1. Very annoying

3. Basic impairments affect to end-user

1. Amplitude distortion

2. Phase distortion

3. Noise

i. Thermal noise

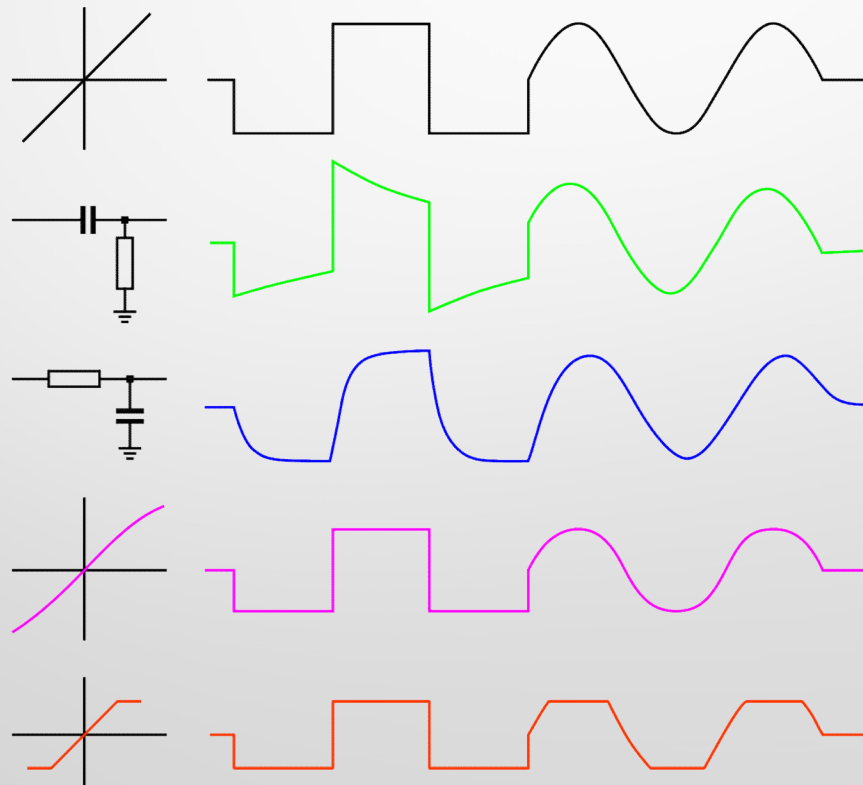
ii. Inter-modulation noise

iii. Impulse noise

iv. Crosstalk noise

Impairments

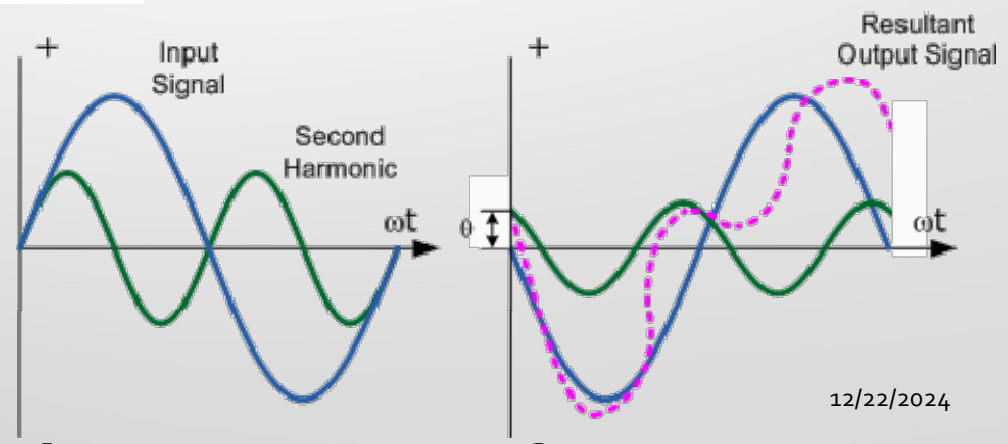
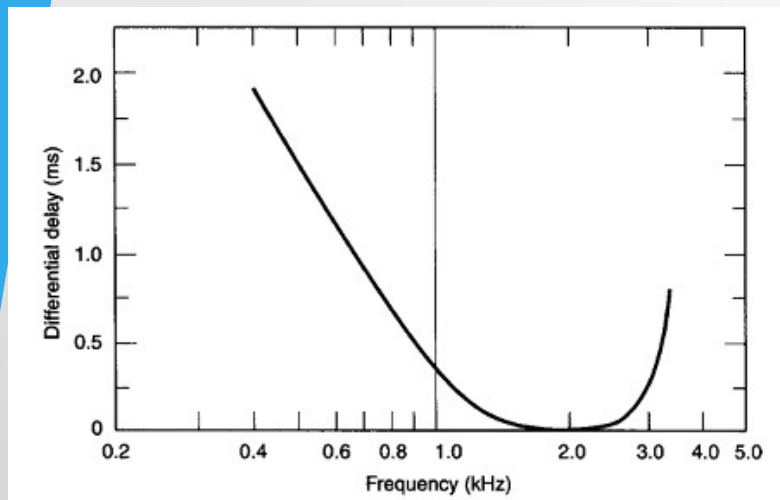
- Amplitude (or attenuation) distortion and Phase distortion
- Noise



Amplitude and phase distortion

- Amplitude distortion:
 - Voice channel occupies the band from 300 to 3400 Hz
 - Some frequencies are attenuated more than others
- ▶ Phase distortion
 - ▶ Velocity of propagation varies with frequency
 - ▶ Filter: increase toward band center and decrease toward band edge

Amplitude and phase distortion



Noise

- Thermal noise
- Intermodulation noise
- Impulse noise
- Crosstalk

Thermal noise

- Caused by random electron motion and characterized by a uniform distribution of energy over the frequency spectrum with a Gaussian distribution of levels
- Thermal noise level (P_n) in 1 Hz bandwidth at absolute zero (0°K)
 - $P_n = -228.6$ dBW per Hz for perfect receiver
- At Temperature room : 290°K (17°C)
 - $P_n = -204$ dBW per Hz = -174 dBm/Hz for perfect receiver

Noise figure

- The difference between the noise output of the actual receiver to the noise output of an perfect receiver with the same overall gain and bandwidth
- Generally, thermal noise level can be given by

$$P_n = -204 \text{ dBW/Hz} + NF_{\text{dB}} + 10 \log B$$

B : bandwidth (Hz)



Noise calculation example

Example: Determine the thermal noise level in dBW of a receiver with 3-dB noise figure and 10-Mhz bandwidth



What will be the thermal noise level of a receiver with a noise figure of 5 dB and a bandwidth of 15 MHz

Assignments

- What will be the thermal noise level of a receiver with a noise figure of 3 dB and a bandwidth of 1 MHz?

Intermodulation (IM) noise

- Two signals with frequencies F_1 and F_2 are passed through a nonlinear device or medium, the result will contain IM products that are spurious frequency energy components.

Impulse noise

- Is noncontinuous, consisting of irregular pulses or noise spikes of short duration and of relative high amplitude?
- Caused by lightning, car ignitions, mechanical switches,...

Crosstalk

- Unwanted coupling between signal paths.
- Caused by:
 - Electrical coupling between transmission media, such as between wire pairs on a voice-frequency (VF) cable system and on digital (PCM) cable systems.
 - Poor control of frequency response
 - Nonlinear performance in analog (FDM) multiplex systems
- Types:
 - Intelligible (4 words are intelligible to the listener from extraneous conversation(s) in a 7-second period)
 - Unintelligible

The received signal from an antenna has maximum amplitude at $1.3\mu V$ and bandwidth 20MHz. It is known that the temperature of environment is $31^{\circ}C$ and the noise figure is 4dB. a) Calculate the SNR_{dB} of the receiver.

b) If the received noise from antenna has amplitude at $0.15\mu V$ (E_N), calculate the SNR_{dB} of the receiver.

c) If the SNR_{dB} requirement of the receiver is 9dB. Do you think that the receiver has a good quality? Why?

An antenna receiver has the received signal with maximum amplitude at $5\mu\text{V}$ and bandwidth 20MHz . It is known that the temperature of environment is 28°C , the noise figure is 4dB and the received noise from antenna has amplitude at $1\mu\text{V}$ (E_N).

- a) Calculate the SNR_{dB} of the receiver.
- b) If the required SNR_{dB} of the receiver is 10dB . Do you think that the receiver has a good quality? Why?

An antenna receiver has the received signal with maximum amplitude at $2.5\mu\text{V}$, the received noise has maximum amplitude at $0.5\mu\text{V}$, bandwidth of the received signal is 20MHz and the noise figure is 4dB . It is assumed that the temperature of environment is 38°C .

- a) Calculate the SNR_{dB} of the receiver.
- b) If the required SNR_{dB} of the receiver is 9dB . Do you think that the receiver has a good quality? Why?

Problems

1. What are the impairments to QoS in a telecommunication system?
2. The noise level of a certain voice channel is measured at -39 dBm, and the test-tone signal level is measured at $+3$ dBm. What is the channel S/N?
3. Explain the cause of phase distortion.
4. Name the four types of noise we are likely to encounter in a telecommunication system.
5. What will be the thermal noise level of a receiver with a noise figure of 3 dB and a bandwidth of 1 MHz? It is assumed that the thermal noise level is -204 dBW/Hz.
6. Calculate SNR dB of a RF receiver with the received signal amplitude is $20\mu\text{V}$ and Bandwidth is 200KHz . It is assumed that noise figure of the system is 4 dB and the thermal noise level is -204 dBW/Hz.